

Master's Comprehensive Examination

Part II

Applied (90 minutes)

November 20, 2004

Instructions: Answer all four questions. Open book and open notes.

Question 1

A chemical plant is suspected of illegal dumping in a nearby river. Samples are taken upstream and downstream from the plant for 12 weeks, and the concentration of a certain chemical is measured. The results are summarized in the table below. Would you conclude that the chemical

Upstream	4.1	1.5	1.0	2.2	2.1	1.2	2.8	4.0	3.7	3.0	5.6	3.3
Downstream	4.5	2.1	1.9	2.0	2.5	1.9	3.4	4.2	3.7	2.9	5.9	3.8

plant is dumping chemicals in the river based on these data? State your assumptions and make sure to give a complete answer.

Question 2

A company that manufactures light bulbs claims their product has an average lifetime of 1000 hours. To test this claim a consumer buys 100 lightbulbs and records the lifetime of each lightbulb. The average lifetime for this sample was 953.2 hours. You can assume lifetime is exponentially distributed, and use appropriate approximations to answer the questions below.

a) Test the hypothesis that the mean lifetime is 1000 hours against the hypothesis that it is less than 1000, at the 5% level.

- b) How big does the sample have to be to detect a lifetime of 950 hours at the 5% level?
- c) What is the power of the test in a) if the mean life time is in fact 700 hours?
- d) What is the power of the test in a) if the mean life time is in fact 1300 hours?
- e) Comment on the sampling procedure used.